GlucoTwin

A Digital Twin for Type 2 Diabetes management



Happiest Health Digital Twin Challenge 2026

Team CodeNCoffee

Prasad V. Potluri Siddhartha Institute of Technology

The Problem

Diabetes care today is a snapshot, not a movie

101 M

people live with diabetes in India (ICMR-INDIAB, 2023)

3 mo

typical gap between HbA1c checks, so daily spikes go unseen

0

safe ways to test a diet or dose change before prescribing it

Post-meal spikes quietly drive neuropathy, nephropathy, retinopathy and heart disease.

Our Solution

EHR history + wearable stream = a living virtual patient



Fuse

Static EHR and 7 days of glucose, heart rate, steps and sleep, combined into one patient model.



Predict

A glucose spike above 180 mg/dL, 2 hours ahead, with the top reasons behind every alert.



Simulate

Test diet, medication and walking changes on the twin before they reach the patient.

Built with **Python** **XGBoost** **scikit-learn** **Streamlit** **Plotly**

Healthcare Use Case

Type 2 Diabetes: continuous glucose and metabolic management

CONDITION

Type 2 Diabetes: glucose and metabolic management

ADVERSE EVENT

Glucose above 180 mg/dL, predicted 120 minutes ahead

WHY 2 HOURS

Time to act: a walk, a lighter meal, a dose, a call

PRIMARY USERS

Diabetologists and physicians; care coaches

Built for Indian patients

- 15 cities across South, West, North and East India
- Rice-based, roti-based and mixed diets, with festival-meal surges
- Early-onset phenotype and high family-history burden
- Sedentary, light and active lifestyles

Two Fused Data Streams

Synthetic, anonymised, and shaped like Indian patients

Static EHR

- Age, sex, city and state
- BMI, HbA1c, baseline blood pressure
- Family history and past diagnoses
- Medication, diet pattern, activity level

+

Dynamic wearable data

- CGM glucose every 15 minutes
- Heart rate and step count
- Sleep quality index (0 to 100)
- Meal log with carbohydrate grams

100
synthetic patients

67,200
wearable readings

7 days
at 15-minute resolution

15
Indian cities

System Architecture

From raw data to a doctor-ready decision, in four steps

1 Data

- Static EHR
- Wearable stream
- Synthetic generator

2 Fusion

- 42 features
- Lags and trends
- 1 h, 3 h, 24 h aggregates
- Meal and sleep context

3 Engines

XGBoost risk engine

2-hour spike alert

Mechanistic twin

What-if simulation

4 Dashboard

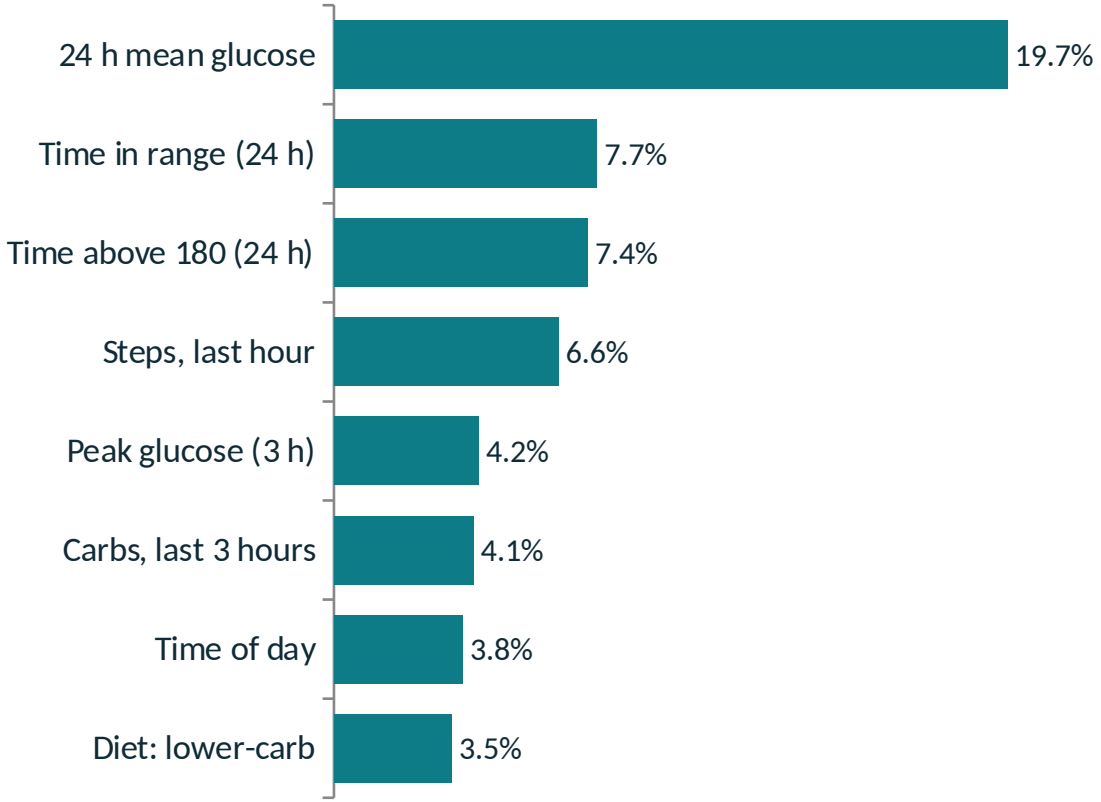
- Ranked patient list
- Explainable alerts
- What-if scenarios
- Live replay clock

Hybrid twin: data-driven accuracy for alerts, physiological reasoning for what-if scenarios

ML Prediction Engine

42 fused features, two XGBoost models, honest validation

What drives the spike prediction



1

Fuse

Glucose lags and trends, 24 h aggregates, heart rate, steps, sleep, meals and EHR in one table.

2

Train

XGBoost classifier and regressor with a patient-level train, validation and test split.

3

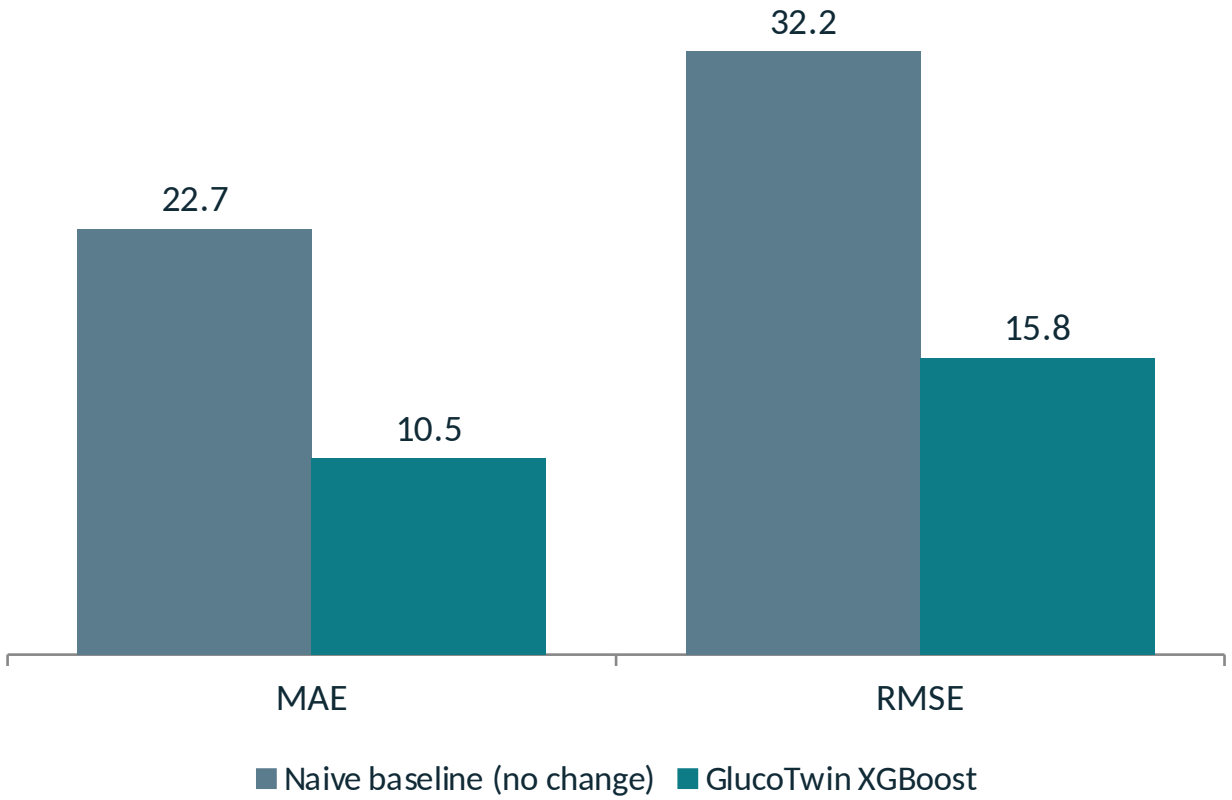
Explain

Every alert lists its top drivers, so the doctor sees why the model is worried.

Results

About half the error of a naive forecast, tested on unseen patients

Error in glucose at +120 min (mg/dL, lower is better)



0.815
F1-score (spike)

93.9%
Accuracy

0.974
ROC-AUC

78.5%
Sensitivity

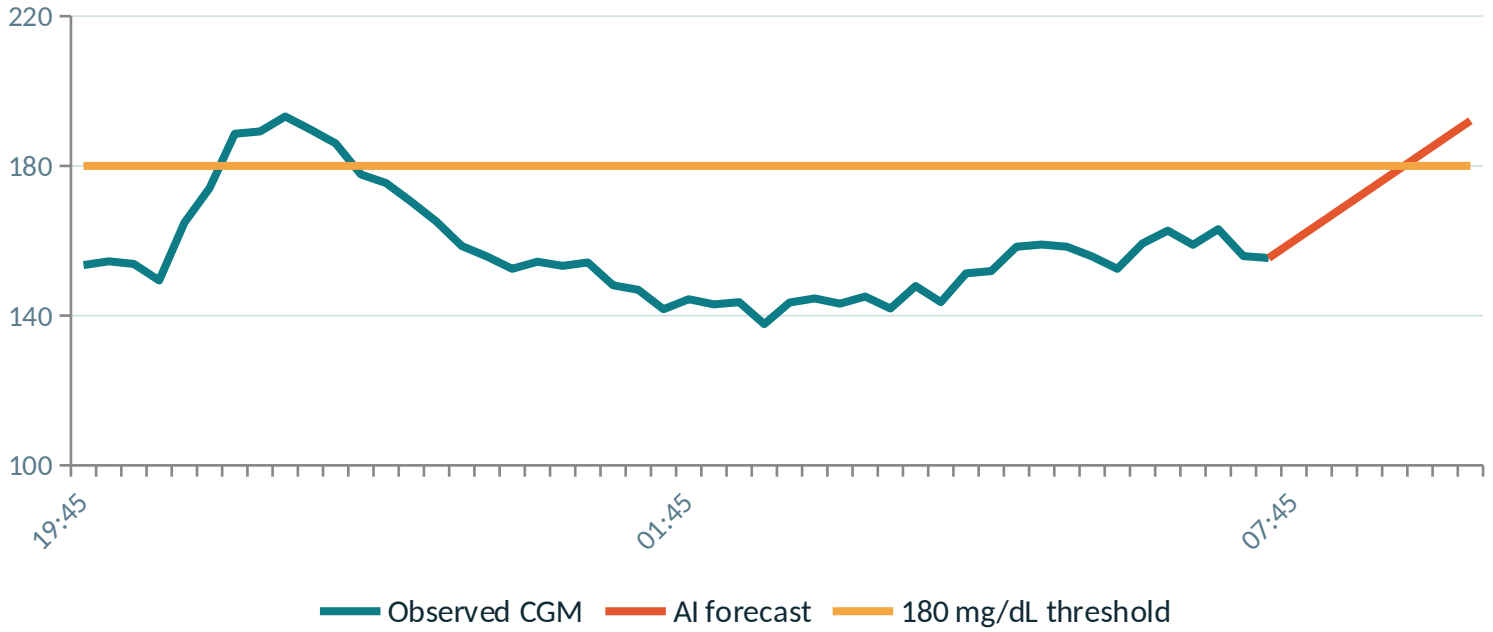
Synthetic data: shows the pipeline works, not clinical validity.

Doctor Dashboard

A real-time alert, with the reasoning behind it

High Risk of Glucose Spike in 120 mins predicted 192 mg/dL by 09:30 | risk 92%

CGM glucose (mg/dL), last 12 h and 2-hour forecast



Patient DT-0032

- 50 y, Male, Chennai
- HbA1c 9.4% | BMI 35.6
- Metformin + Basal Insulin
- South Indian rice-based diet

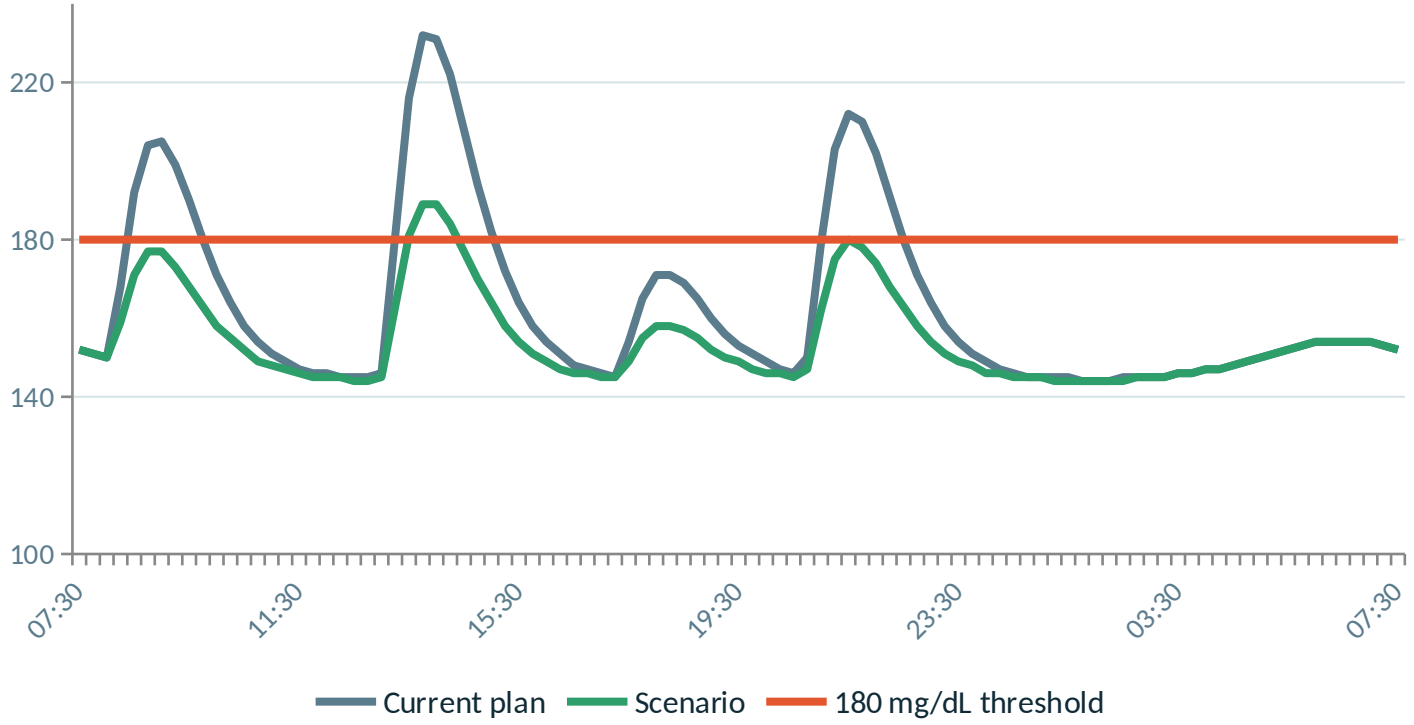
Tabs: cohort overview, digital twin, what-if, model and methods

Replay clock streams data in 15-minute steps

What-If Simulator

Test a plan on the twin before prescribing it

Next 24 hours of glucose (mg/dL)



Twin calibrated to this patient's last 3 days of CGM (RMSE 8.2 mg/dL). Effect sizes are illustrative.

SCENARIO

Carbs at 70% of usual, plus a 20-minute walk after each meal

Time above 180 mg/dL

4.8 h → 1.0 h

Peak glucose

232 → 189 mg/dL

Time in range (70 to 180)

80% → 96%

Impact, Safety and Roadmap

Useful today as a prototype, designed to grow into a clinical tool



Impact

- Act before spikes happen
- Fewer long-term complications
- Scalable follow-up between visits



Safety and ethics

- 100% synthetic, anonymised data
- Decision support, not decision making
- Explainable alerts, tested on unseen patients



Roadmap

- Real CGM and wearable ingestion
- FHIR and ABDM integration
- Clinical validation and a multilingual patient app

Thank you

Questions?

Team CodeNCoffee

Prasad V. Potluri Siddhartha Institute of Technology

GitHub: [repository link]

Demo video: [unlisted YouTube link]

Open source under the MIT License

